

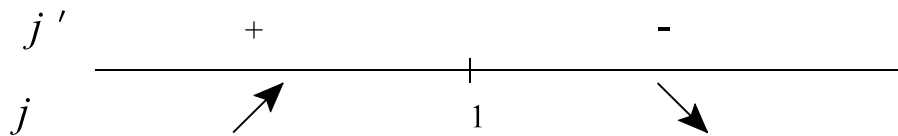
$$j(x) = 4xe^{-x} + 1$$

a. Domain = $\mathbb{R}$

b. Intercepts $(0,1), (-0.2039,0)$

c. $j'(x) = 4e^{-x} - 4xe^{-x} = 4e^{-x}(1-x) \Rightarrow$ C.N. $x = 1$

d.



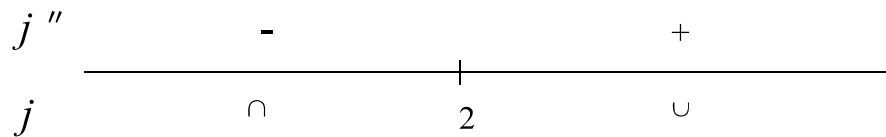
j is decreasing on $(1, \infty)$

j is increasing on $(-\infty, 1)$

e. $P(1, 1 + 4e^{-1})$ is a relative maximum point on the graph

f. $j''(x) = -8e^{-x} + 4xe^{-x} = 4e^{-x}(x-2) \Rightarrow$ 2nd C.N. $x = 2$

g.



j is concave up on $(2, \infty)$

j is concave down on $(-\infty, 2)$

h. Inflection point $P(2, 1 + 8e^{-2})$

i.

j.

-4, -872.571

-3, -240.026

-2, -58.1130

-1, -9.87300

0, 1.00000

1, 2.47200

2, 2.08300

3, 1.59700

4, 1.29300

